Digital Signal Processing - Lecture Notes

1. Discrete Signals and the Z-Transform

A discrete-time signal x[n] is defined at integer n. The Z-transform X(z) = sum over n of x[n] z^(-n). The region of convergence determines uniqueness. The Z-transform converts convolution into multiplication: if y[n] = x[n] * h[n] then Y(z) = X(z)H(z), and the transfer function is H(z).

A causal system has a right-sided impulse response. For a stable causal system all poles of H(z) lie inside the unit circle |z| = 1.

2. The Discrete Fourier Transform

The N-point Discrete Fourier Transform is X[k] = sum from n=0 to N-1 of x[n] e^(-j*2*pi*k*n/N), for k = 0 to N-1. The DFT is periodic with period N and relates to the DFT of a signal sampled at fs, where bin k corresponds to frequency k*fs/N.

The frequency resolution of an N-point DFT is fs/N. Zero-padding interpolates the spectrum without increasing the true resolution. Spectral leakage occurs because a finite record truncates the signal; windowing (Hamming, Hann, Blackman) reduces leakage at the cost of widening the main lobe.

3. The Fast Fourier Transform

The Fast Fourier Transform (FFT) computes the DFT with complexity O(N log N) instead of the direct O(N^2). The radix-2 Cooley-Tukey algorithm requires N to be a power of two and splits the transform into even and odd indexed sub-transforms recursively.

For N = 1024 the FFT needs about 10*N operations instead of N^2, roughly a hundred-fold saving.

4. Digital Filters

Finite Impulse Response (FIR) filters have an impulse response of finite length, are always stable and can be designed with exactly linear phase. Infinite Impulse Response (IIR) filters use feedback, need fewer coefficients for the same magnitude response but have non-linear phase and can be unstable.

A moving-average filter of length M is y[n] = (1/M) * sum of the last M inputs; it is a simple low-pass FIR filter. The bilinear transform maps an analogue prototype filter to the digital domain and is used to design IIR filters such as Butterworth and Chebyshev filters.